

DESERTO DE ATACMA, Chile

WMO: 854670

Lat: 27.264S

Lon: 70.774W

Elev: 210

StdP: 98.83

Time Zone: -4.00 (CHL)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.9	7.0	1.8	4.4	9.9	3.2	4.9	9.6	8.9	14.4	8.1	14.5	2.4	100	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.7	25.1	17.5	24.1	17.2	23.2	16.8	18.2	23.7	17.7	23.0	17.1	22.4	5.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.0	11.7	20.5	15.2	11.1	19.8	14.9	10.9	19.6	52.4	23.8	50.5	23.3	49.0	22.6	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.3	6.6	3.6	27.9	0.8	1.6	3.0	29.1	2.5	30.0	2.0	30.9	1.4	32.0		
(5)			2.2	19.2	1.0	1.0	1.4	19.9	0.9	20.5	0.3	21.0	-0.4	21.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	19.7	19.8	18.5	16.2	14.5	12.9	12.3	12.9	13.5	14.7	16.2	17.8	(6)
(7)		DBStd	2.91	1.33	1.28	1.42	1.53	1.52	1.51	1.44	1.48	1.38	1.20	1.14	1.24	(7)
(8)		HDD10.0	2	0	0	0	0	0	1	1	1	0	0	0	0	(8)
(9)		HDD18.3	1071	3	2	16	67	119	163	187	169	144	112	64	26	(9)
(10)		CDD10.0	2091	299	276	262	186	140	87	73	89	106	146	187	241	(10)
(11)		CDD18.3	118	44	43	19	2	1	0	0	0	0	0	1	8	(11)
(12)		CDH23.3	197	78	81	23	4	2	1	1	3	0	0	1	3	(12)
(13)		CDH26.7	7	3	4	0	1	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	3.3	3.4	3.4	3.3	3.1	2.8	2.9	3.0	3.3	3.4	3.7	3.7	3.7	(14)
(15)	Precipitation	PrecAvg	26	0	0	1	1	3	10	5	5	0	0	0	0	(15)
(16)		PrecMax	123	1	0	4	16	26	60	20	61	1	2	0	3	(16)
(17)		PrecMin	2	0	0	0	0	0	0	1	0	0	0	0	0	(17)
(18)		PrecStd	33	0	0	1	4	6	19	5	14	0	0	0	1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.0	27.1	25.1	23.8	22.2	21.1	20.2	22.0	20.9	21.1	22.2	23.9	(19)
(20)		MCWB	18.6	18.6	17.5	16.3	14.7	12.6	12.6	12.2	14.0	14.6	15.2	16.6	16.6	(20)
(21)		2%	DB	25.1	25.1	24.0	22.0	20.1	18.8	18.2	18.8	19.0	19.9	21.1	22.9	(21)
(22)		MCWB	17.6	17.7	17.1	15.5	14.4	12.7	12.6	12.4	13.4	14.0	14.9	16.0	16.0	(22)
(23)		5%	DB	24.1	24.1	22.9	20.9	18.9	17.2	16.9	17.1	17.9	18.9	20.2	22.0	(23)
(24)		MCWB	17.3	17.5	16.7	15.3	13.8	12.3	12.1	12.1	12.7	13.5	14.5	15.6	15.6	(24)
(25)		10%	DB	23.1	23.2	21.9	19.8	17.8	16.1	15.8	16.0	16.8	17.9	19.2	21.1	(25)
(26)		MCWB	16.9	17.0	16.3	14.8	13.3	12.2	11.8	11.9	12.2	13.0	14.1	15.5	15.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.5	19.1	18.4	16.8	16.0	14.2	14.1	13.9	14.5	14.8	15.7	17.1	(27)
(28)		MCDWB	24.6	25.6	23.8	22.0	20.8	18.8	18.2	19.7	19.7	20.1	21.1	22.6	22.6	(28)
(29)		2%	WB	18.4	18.2	17.4	16.0	14.9	13.4	13.2	13.1	13.7	14.1	15.2	16.4	(29)
(30)		MCDWB	23.8	23.9	22.5	20.6	19.1	17.4	17.0	17.4	18.2	19.0	20.4	21.6	21.6	(30)
(31)		5%	WB	17.6	17.8	16.9	15.4	14.2	12.8	12.5	12.5	13.0	13.7	14.6	15.9	(31)
(32)		MCDWB	23.1	23.2	21.8	19.7	17.8	16.4	16.0	16.3	16.9	18.1	19.5	20.9	20.9	(32)
(33)		10%	WB	17.1	17.2	16.4	14.8	13.7	12.2	11.9	12.0	12.4	13.2	14.2	15.5	(33)
(34)		MCDWB	22.5	22.4	20.9	18.6	16.9	15.4	15.0	15.4	16.0	17.2	18.6	20.2	20.2	(34)
(35)	Mean Daily Temperature Range	MDBR	7.3	7.7	7.2	7.6	7.3	7.9	8.0	7.7	7.6	7.2	7.1	7.2	7.2	(35)
(36)		5% DB	MCDBR	8.3	8.5	8.3	9.2	9.2	9.7	10.1	9.8	9.1	8.6	8.0	8.1	(36)
(37)		MCWBR	3.2	3.5	3.5	4.4	4.9	5.4	5.6	5.1	4.6	4.2	3.5	3.3	3.3	(37)
(38)		5% WB	MCDWB	7.7	7.9	7.8	8.4	8.5	8.7	9.2	8.8	8.3	8.1	7.6	7.5	(38)
(39)		MCWBR	3.1	3.3	3.4	4.1	4.7	5.1	5.3	4.8	4.3	4.0	3.4	3.4	3.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.369	0.362	0.349	0.345	0.322	0.308	0.304	0.310	0.327	0.325	0.330	0.345	(40)	
(41)		taud	2.378	2.399	2.428	2.415	2.469	2.506	2.502	2.481	2.418	2.440	2.446	2.419	(41)	
(42)		Ebn at Noon	973	963	945	899	877	871	890	925	952	989	1005	999	(42)	
(43)		Edh at Noon	130	124	114	105	90	83	86	97	113	118	121	125	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.79	7.32	6.04	4.64	3.50	3.18	3.47	4.28	5.48	6.53	7.36	7.78	(44)	
(45)		RadStd	0.31	0.28	0.26	0.25	0.24	0.17	0.14	0.22	0.20	0.29	0.27	0.41	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page